

Functional Programming

Data Wrangling in R

Functional Programming

“R, at its heart, is a functional programming (FP) language. This means that it provides many tools for the creation and manipulation of functions.” - [Hadley Wickham](#)

Don't need to write for-loops! - check this [video](#).

Iterative work

Allows you to flexibly iterate functions to multiple elements of a data object!

Useful when you want to apply a function to:

- * lots of columns in a tibble
- * multiple tibbles
- * multiple data files
- * or perform fancy functions with vectors (or tibble columns)

base R **apply** functions

Works really simply for all columns.

```
sapply(mtcars, FUN = round)
```

##		mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
##	[1,]	21	6	160	110	4	3	16	0	1	4	4
##	[2,]	21	6	160	110	4	3	17	0	1	4	4
##	[3,]	23	4	108	93	4	2	19	1	1	4	1
##	[4,]	21	6	258	110	3	3	19	1	0	3	1
##	[5,]	19	8	360	175	3	3	17	0	0	3	2
##	[6,]	18	6	225	105	3	3	20	1	0	3	1
##	[7,]	14	8	360	245	3	4	16	0	0	3	4
##	[8,]	24	4	147	62	4	3	20	1	0	4	2
##	[9,]	23	4	141	95	4	3	23	1	0	4	2
##	[10,]	19	6	168	123	4	3	18	1	0	4	4
##	[11,]	18	6	168	123	4	3	19	1	0	4	4
##	[12,]	16	8	276	180	3	4	17	0	0	3	3
##	[13,]	17	8	276	180	3	4	18	0	0	3	3
##	[14,]	15	8	276	180	3	4	18	0	0	3	3
##	[15,]	10	8	472	205	3	5	18	0	0	3	4
##	[16,]	10	8	460	215	3	5	18	0	0	3	4
##	[17,]	15	8	440	230	3	5	17	0	0	3	4
##	[18,]	32	4	79	66	4	2	19	1	1	4	1
##	[19,]	30	4	76	52	5	2	19	1	1	4	2
##	[20,]	34	4	71	65	4	2	20	1	1	4	1
##	[21,]	22	4	120	97	4	2	20	1	0	3	1
##	[22,]	16	8	318	150	3	4	17	0	0	3	2
##	[23,]	15	8	304	150	3	3	17	0	0	3	2

Base R **apply** functions specific columns

Harder to use for specific columns.

```
rounded_mtcars <- mtcars %>% select(dispatch, drat, wt, qsec) %>%  
  apply(FUN = round)  
bind_cols(rounded_mtcars, select(mtcars, -c(dispatch, drat, wt, qsec)))
```

	dispatch	drat	wt	qsec	mpg	cyl	hp	vs	am	gear	carb
## Mazda RX4	160	4	3	16	21.0	6	110	0	1	4	4
## Mazda RX4 Wag	160	4	3	17	21.0	6	110	0	1	4	4
## Datsun 710	108	4	2	19	22.8	4	93	1	1	4	1
## Hornet 4 Drive	258	3	3	19	21.4	6	110	1	0	3	1
## Hornet Sportabout	360	3	3	17	18.7	8	175	0	0	3	2
## Valiant	225	3	3	20	18.1	6	105	1	0	3	1
## Duster 360	360	3	4	16	14.3	8	245	0	0	3	4
## Merc 240D	147	4	3	20	24.4	4	62	1	0	4	2
## Merc 230	141	4	3	23	22.8	4	95	1	0	4	2
## Merc 280	168	4	3	18	19.2	6	123	1	0	4	4
## Merc 280C	168	4	3	19	17.8	6	123	1	0	4	4
## Merc 450SE	276	3	4	17	16.4	8	180	0	0	3	3
## Merc 450SL	276	3	4	18	17.3	8	180	0	0	3	3
## Merc 450SLC	276	3	4	18	15.2	8	180	0	0	3	3
## Cadillac Fleetwood	472	3	5	18	10.4	8	205	0	0	3	4
## Lincoln Continental	460	3	5	18	10.4	8	215	0	0	3	4
## Chrysler Imperial	440	3	5	17	14.7	8	230	0	0	3	4
## Fiat 128	79	4	2	19	32.4	4	66	1	1	4	1
## Honda Civic	76	5	2	19	30.4	4	52	1	1	4	2
## Toyota Corolla	71	4	2	20	33.9	4	65	1	1	4	1
## Toyota Corona	120	4	2	20	21.5	4	97	1	0	3	1

Working **across** multiple columns

Say we wanted to round multiple columns of the `mtcars` data. We could do so one column at a time, or we could use the `across` function from the `dplyr` package. Needs to be used **within other dplyr functions** such as `mutate`.

```
mutate(across(which_columns, which function or operation))
```

```
head(mtcars, 2)
```

```
##           mpg  cyl  disp  hp  drat    wt   qsec  vs  am  gear  carb
## Mazda RX4    21   6  160  110   3.9 2.620  16.46  0  1     4     4
## Mazda RX4 Wag 21   6  160  110   3.9 2.875  17.02  0  1     4     4
```

```
mtcars %>%
  mutate(across(.cols = c(disp, drat, wt, qsec), round)) %>%
  head(2)
```

```
##           mpg  cyl  disp  hp  drat    wt   qsec  vs  am  gear  carb
## Mazda RX4    21   6  160  110    4  3    16  0  1     4     4
## Mazda RX4 Wag 21   6  160  110    4  3    17  0  1     4     4
```

Special tilde use

If you see `~ .x` or `. this means we are passing x to a function.`

See [purrr-shortcuts](#) See here for info about the [difference between them](#).

For example - this is not necessary but you could use it here:

```
mtcars %>%  
  mutate(across(.cols = c(displ, drat, wt, qsec), ~ round(.x))) %>%  
  head(2)
```

```
##           mpg cyl displ  hp drat   wt  qsec vs am gear carb  
## Mazda RX4      21   6  160 110   4.35 16.99   0  1   4     4  
## Mazda RX4 Wag  21   6  160 110   4.35 17.05   0  1   4     4
```

```
mtcars %>%  
  mutate(across(.cols = c(displ, drat, wt, qsec), ~ round(.))) %>%  
  head(2)
```

```
##           mpg cyl displ  hp drat   wt  qsec vs am gear carb  
## Mazda RX4      21   6  160 110   4.35 16.99   0  1   4     4  
## Mazda RX4 Wag  21   6  160 110   4.35 17.05   0  1   4     4
```

Using **across** with arguments

If you wish to also pass arguments to the function that you are applying to the various columns, then you need to use the `~` and `.x` (or `.`) as a place holder for what you the values you will be passing into the function.

```
mtcars %>%  
  mutate(across(.cols = c(displ, drat, wt, qsec), ~ round(.x, digits = 1))) %>%  
  head(n = 2)
```

```
##           mpg cyl displ  hp drat   wt  qsec vs am gear carb  
## Mazda RX4      21   6  160 110   3.9 2.6 16.5  0  1    4    4  
## Mazda RX4 Wag  21   6  160 110   3.9 2.9 17.0  0  1    4    4
```

```
mtcars %>%  
  mutate(across(.cols = c(displ, drat, wt, qsec), ~ round(., digits = 1))) %>%  
  head(n = 2)
```

```
##           mpg cyl displ  hp drat   wt  qsec vs am gear carb  
## Mazda RX4      21   6  160 110   3.9 2.6 16.5  0  1    4    4  
## Mazda RX4 Wag  21   6  160 110   3.9 2.9 17.0  0  1    4    4
```


Previously we filtered for patterns or conditions..

Dilemma: Seems a bit repetitive!

```
mtcars %>%  
  filter(cyl > 3 & cyl < 8,  
         gear > 3 & gear < 8,  
         carb > 3 & carb < 8)
```

##		mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
##	Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4
##	Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4
##	Merc 280	19.2	6	167.6	123	3.92	3.440	18.30	1	0	4	4
##	Merc 280C	17.8	6	167.6	123	3.92	3.440	18.90	1	0	4	4
##	Ferrari Dino	19.7	6	145.0	175	3.62	2.770	15.50	0	1	5	6

Now we can filter multiple columns!

`if_all()`: helps us filter on multiple similar conditions simultaneously!

```
mtcars %>%  
  filter(if_all(c(cyl, gear, carb), ~.x > 3 & .x < 8))
```

##		mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
##	Mazda RX4	21.0	6	160.0	110	3.90	2.620	16.46	0	1	4	4
##	Mazda RX4 Wag	21.0	6	160.0	110	3.90	2.875	17.02	0	1	4	4
##	Merc 280	19.2	6	167.6	123	3.92	3.440	18.30	1	0	4	4
##	Merc 280C	17.8	6	167.6	123	3.92	3.440	18.90	1	0	4	4
##	Ferrari Dino	19.7	6	145.0	175	3.62	2.770	15.50	0	1	5	6

Writing functions

Why write your own functions?

- Cut down on repetitive code (easier to fix things!)
- Organize code into manageable chunks
- Avoid running code unintentionally

Functions in R

To create a function we can use the `function` function and specify what input the function will take and what it will do to it.

```
my_function <- function(x){x + 1}  
my_function
```

```
## function(x){x + 1}  
## <environment: 0x1400f2560>
```

```
my_data <- c(2,3,4)  
my_function(x = my_data)
```

```
## [1] 3 4 5
```

```
my_function(my_data)
```

```
## [1] 3 4 5
```

shortcut for functions

Alternatively we can use `\(x)`. See this [link about function shortcuts](#).

```
my_function <- \(x){x + 1}  
my_function
```

```
## \(x){x + 1}  
## <environment: 0x1400f2560>
```

```
my_function(x = my_data)
```

```
## [1] 3 4 5
```

purrr is also a super helpful package!

“Designed to make your functions purrr.”

`dplyr` is designed for data frames `purrr` is designed for vectors

The `purrr` package can be very helpful!

- <https://purrr.tidyverse.org/>
- <https://github.com/rstudio/cheatsheets/raw/master/purrr.pdf>
- <https://jennybc.github.io/purrr-tutorial/>

purrr main functions

map and map_* and modify

- **applies function** to each element of an vector or object (map returns a list, modify returns the same object type) - map_df will output a dataframe

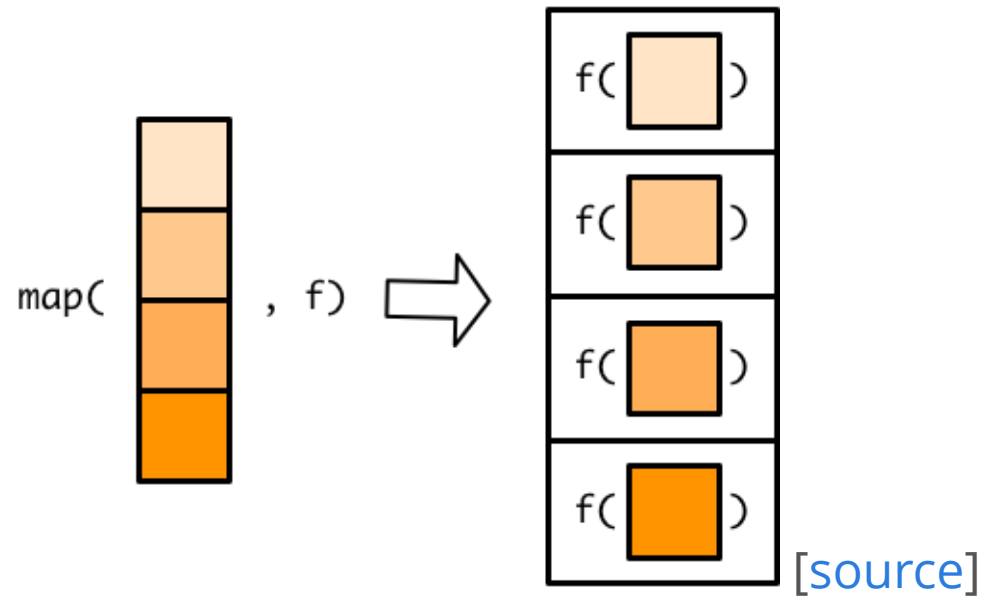
map2 and map2_*

- applies function to each element of **two** vectors or objects

pmap and pmap_* - applies function to each element of **3+** vector or objects (requires a list for input)

the *_* options specify the type of data output

Using `map()`



Using `map_df()` and `modify()`

```
my_tibble <- tibble(values = c(1.2, 2.3, 3.5, 4.6))  
map_df(my_tibble, round)
```

```
## # A tibble: 4 × 1  
##   values  
##   <dbl>  
## 1     1  
## 2     2  
## 3     4  
## 4     5
```

```
modify(my_tibble, round)
```

```
## # A tibble: 4 × 1  
##   values  
##   <dbl>  
## 1     1  
## 2     2  
## 3     4  
## 4     5
```

It's a bit simpler than across...

But `across` keeps rownames.

```
mtcars %>%  
  modify(format, digits = 1) %>%  
  head(n = 2)
```

```
##      mpg cyl disp  hp drat  wt  qsec vs am gear carb  
## 1   21   6  160 110   4   3    16  0  1    4    4  
## 2   21   6  160 110   4   3    17  0  1    4    4
```

```
mtcars %>%  
  mutate(across(.cols = everything(), ~ format(.x, digits = 1))) %>%  
  head(n = 2)
```

```
##  
## Mazda RX4      21   6  160 110   4   3    16  0  1    4    4  
## Mazda RX4 Wag  21   6  160 110   4   3    17  0  1    4    4
```

purrr apply function to some columns like across

Using `modify_if()` (or `map_if()`), we can specify what columns to modify

```
head(as_tibble(iris), 3)
```

```
## # A tibble: 3 × 5
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##   <dbl>         <dbl>         <dbl>         <dbl> <fct>
## 1      5.1         3.5           1.4           0.2 setosa
## 2      4.9         3             1.4           0.2 setosa
## 3      4.7         3.2           1.3           0.2 setosa
```

```
as_tibble(iris) %>%
  modify_if(is.numeric, as.character) %>%
  head(3)
```

```
## # A tibble: 3 × 5
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##   <chr>         <chr>         <chr>         <chr> <fct>
## 1 5.1         3.5           1.4           0.2 setosa
## 2 4.9         3             1.4           0.2 setosa
## 3 4.7         3.2           1.3           0.2 setosa
```

Lists

What is a 'list'?

- Lists are the most flexible/"generic" data class in R
- Can be created using `list()`
- Can hold vectors, strings, matrices, models, list of other lists, lists upon lists!
- Can reference data using `$` (if the elements are named), or using `[]`, or `[[ ]]`

```
mylist <- list(  
  letters = c("A", "b", "c"),  
  numbers = 1:3,  
  matrix(1:25, ncol = 5),  
  matrix(1:25, ncol = 5)  
)
```

List Structure

```
head(mylist)
```

```
## $letters
## [1] "A" "b" "c"
##
## $numbers
## [1] 1 2 3
##
## [[3]]
##      [,1] [,2] [,3] [,4] [,5]
## [1,]     1     6    11    16    21
## [2,]     2     7    12    17    22
## [3,]     3     8    13    18    23
## [4,]     4     9    14    19    24
## [5,]     5    10    15    20    25
##
## [[4]]
##      [,1] [,2] [,3] [,4] [,5]
## [1,]     1     6    11    16    21
## [2,]     2     7    12    17    22
## [3,]     3     8    13    18    23
## [4,]     4     9    14    19    24
## [5,]     5    10    15    20    25
```

List referencing

```
mylist[1] # returns a list
```

```
## $letters  
## [1] "A" "b" "c"
```

```
mylist["letters"] # returns a list
```

```
## $letters  
## [1] "A" "b" "c"
```


List referencing

```
mylist[[1]] # returns the vector 'letters'
```

```
## [1] "A" "b" "c"
```

```
mylist$letters # returns vector
```

```
## [1] "A" "b" "c"
```

```
mylist[["letters"]] # returns the vector 'letters'
```

```
## [1] "A" "b" "c"
```

List referencing

You can also select multiple lists with the single brackets.

```
mylist[1:2] # returns a list
```

```
## $letters  
## [1] "A" "b" "c"  
##  
## $numbers  
## [1] 1 2 3
```

... Why lists?

List: `split()` a dataset

We can create a list by splitting up a dataframe. We will use `mtcars`.

```
head(mtcars)
```

##		mpg	cyl	disp	hp	drat	wt	qsec	vs	am	gear	carb
##	Mazda RX4	21.0	6	160	110	3.90	2.620	16.46	0	1	4	4
##	Mazda RX4 Wag	21.0	6	160	110	3.90	2.875	17.02	0	1	4	4
##	Datsun 710	22.8	4	108	93	3.85	2.320	18.61	1	1	4	1
##	Hornet 4 Drive	21.4	6	258	110	3.08	3.215	19.44	1	0	3	1
##	Hornet Sportabout	18.7	8	360	175	3.15	3.440	17.02	0	0	3	2
##	Valiant	18.1	6	225	105	2.76	3.460	20.22	1	0	3	1

List: `split()` the dataset by `cyl`

```
mtcars_split <-mtcars %>% split(.$cyl)
str(mtcars_split)
```

```
## List of 3
## $ 4:'data.frame':  11 obs. of  11 variables:
##   ..$ mpg : num [1:11] 22.8 24.4 22.8 32.4 30.4 33.9 21.5 27.3 26 30.4 ...
##   ..$ cyl : num [1:11] 4 4 4 4 4 4 4 4 4 4 4 ...
##   ..$ disp: num [1:11] 108 146.7 140.8 78.7 75.7 ...
##   ..$ hp  : num [1:11] 93 62 95 66 52 65 97 66 91 113 ...
##   ..$ drat: num [1:11] 3.85 3.69 3.92 4.08 4.93 4.22 3.7 4.08 4.43 3.77 ...
##   ..$ wt  : num [1:11] 2.32 3.19 3.15 2.2 1.61 ...
##   ..$ qsec: num [1:11] 18.6 20 22.9 19.5 18.5 ...
##   ..$ vs  : num [1:11] 1 1 1 1 1 1 1 1 0 1 ...
##   ..$ am  : num [1:11] 1 0 0 1 1 1 0 1 1 1 ...
##   ..$ gear: num [1:11] 4 4 4 4 4 4 3 4 5 5 ...
##   ..$ carb: num [1:11] 1 2 2 1 2 1 1 1 2 2 ...
## $ 6:'data.frame':  7 obs. of  11 variables:
##   ..$ mpg : num [1:7] 21 21 21.4 18.1 19.2 17.8 19.7
##   ..$ cyl : num [1:7] 6 6 6 6 6 6 6
##   ..$ disp: num [1:7] 160 160 258 225 168 ...
##   ..$ hp  : num [1:7] 110 110 110 105 123 123 175
##   ..$ drat: num [1:7] 3.9 3.9 3.08 2.76 3.92 3.92 3.62
##   ..$ wt  : num [1:7] 2.62 2.88 3.21 3.46 3.44 ...
##   ..$ qsec: num [1:7] 16.5 17 19.4 20.2 18.3 ...
##   ..$ vs  : num [1:7] 0 0 1 1 1 1 0
##   ..$ am  : num [1:7] 1 1 0 0 0 0 1
##   ..$ gear: num [1:7] 4 4 3 3 4 4 5
##   ..$ carb: num [1:7] 4 4 1 1 4 4 6
## $ 8:'data.frame':  14 obs. of  11 variables:
```

List: model on each

```
mtcars %>%  
  split(.$cyl) %>% # creates split of data for each unique cyl value  
  map(~lm(mpg ~ wt, data = .)) %>% # apply linear model to each  
  map(summary) %>%  
  map_dbl("r.squared")
```

```
##           4           6           8  
## 0.5086326 0.4645102 0.4229655
```

Use lists to iterate through multiple files!

This comes up a lot in data cleaning and also when reading in multiple files!

```
library(here)
library(readr)
file_list <- list.files(here::here("data/iris/"), pattern = "*.csv")

file_list <- paste0(here::here("data/iris/"), file_list)
file_list
```

```
## [1] "/Users/avahoffman/Dropbox/JHSPH/Data-Wrangling_SISBID/data/iris/iris_q1.csv"
## [2] "/Users/avahoffman/Dropbox/JHSPH/Data-Wrangling_SISBID/data/iris/iris_q4.csv"
## [3] "/Users/avahoffman/Dropbox/JHSPH/Data-Wrangling_SISBID/data/iris/iris_q5.csv"
```

```
multifile_data <- file_list %>%
  map(read_csv)
```

Reading in multiple files -First item

```
multifile_data[[1]]
```

```
## # A tibble: 150 × 5
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##   <dbl>         <dbl>         <dbl>         <dbl> <chr>
## 1         5.1         3.5         1.4         0.2 setosa
## 2         4.9         3         1.4         0.2 setosa
## 3         4.7         3.2         1.3         0.2 setosa
## 4         4.6         3.1         1.5         0.2 setosa
## 5         5         3.6         1.4         0.2 setosa
## 6         5.4         3.9         1.7         0.4 setosa
## 7         4.6         3.4         1.4         0.3 setosa
## 8         5         3.4         1.5         0.2 setosa
## 9         4.4         2.9         1.4         0.2 setosa
## 10        4.9         3.1         1.5         0.1 setosa
## # i 140 more rows
```


Reading in multiple files - 2nd item

```
multifile_data[[2]]
```

```
## # A tibble: 150 × 1
##   `Sepal.Length:Sepal.Width:Petal.Length:Petal.Width:Species`
##   <chr>
## 1 5.1:3.5:1.4:0.2:setosa
## 2 4.9:3.1:1.4:0.2:setosa
## 3 4.7:3.2:1.3:0.2:setosa
## 4 4.6:3.1:1.5:0.2:setosa
## 5 5.3:6.1:1.4:0.2:setosa
## 6 5.4:3.9:1.7:0.4:setosa
## 7 4.6:3.4:1.4:0.3:setosa
## 8 5.3:4.1:1.5:0.2:setosa
## 9 4.4:2.9:1.4:0.2:setosa
## 10 4.9:3.1:1.5:0.1:setosa
## # i 140 more rows
```

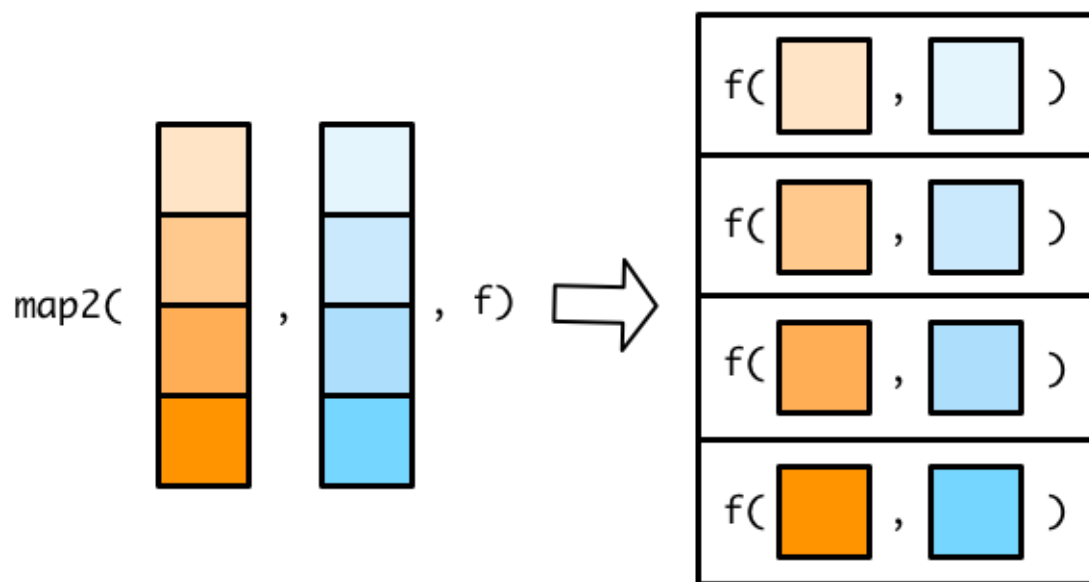
Reading in multiple files - 3rd item

```
multifile_data[[3]]
```

```
## # A tibble: 150 × 5
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##   <dbl>         <dbl>         <dbl>         <dbl> <chr>
## 1      -999          3.5          1.4          0.2 setosa
## 2      -999          3          1.4          0.2 setosa
## 3      -999          3.2          1.3          0.2 setosa
## 4       4.6          3.1          1.5          0.2 setosa
## 5       5           3.6          1.4          0.2 setosa
## 6       5.4          3.9          1.7          0.4 setosa
## 7       4.6          3.4          1.4          0.3 setosa
## 8       5           3.4          1.5          0.2 setosa
## 9       4.4          2.9          1.4          0.2 setosa
## 10      4.9          3.1          1.5          0.1 setosa
## # i 140 more rows
```

Import with different delimiters

```
delimiters <- c(",", ":", ";") # delimiters for each file  
# Write our own function to read files with specific delimiters:  
read_with_delimiter <- function(file, delimiter) {  
  read_delim(file, delim = delimiter)  
}  
  
# Map over file_list and delimiters  
multifile_data <- map2(file_list, delimiters, read_with_delimiter)
```



[\[source\]](#)

Reading in multiple files - 1st item

```
multifile_data[[1]]
```

```
## # A tibble: 150 × 5
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##   <dbl>         <dbl>         <dbl>         <dbl> <chr>
## 1         5.1         3.5         1.4         0.2 setosa
## 2         4.9         3         1.4         0.2 setosa
## 3         4.7         3.2         1.3         0.2 setosa
## 4         4.6         3.1         1.5         0.2 setosa
## 5         5         3.6         1.4         0.2 setosa
## 6         5.4         3.9         1.7         0.4 setosa
## 7         4.6         3.4         1.4         0.3 setosa
## 8         5         3.4         1.5         0.2 setosa
## 9         4.4         2.9         1.4         0.2 setosa
## 10        4.9         3.1         1.5         0.1 setosa
## # i 140 more rows
```

Reading in multiple files - 2nd item

```
multifile_data[[2]]
```

```
## # A tibble: 150 × 5
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##   <dbl>         <dbl>         <dbl>         <dbl> <chr>
## 1         5.1         3.5         1.4         0.2 setosa
## 2         4.9         3         1.4         0.2 setosa
## 3         4.7         3.2         1.3         0.2 setosa
## 4         4.6         3.1         1.5         0.2 setosa
## 5         5         3.6         1.4         0.2 setosa
## 6         5.4         3.9         1.7         0.4 setosa
## 7         4.6         3.4         1.4         0.3 setosa
## 8         5         3.4         1.5         0.2 setosa
## 9         4.4         2.9         1.4         0.2 setosa
## 10        4.9         3.1         1.5         0.1 setosa
## # i 140 more rows
```

Reading in multiple files - 3rd item

```
multifile_data[[3]]
```

```
## # A tibble: 150 × 5
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
##   <dbl>         <dbl>         <dbl>         <dbl> <chr>
## 1      -999          3.5          1.4          0.2 setosa
## 2      -999          3          1.4          0.2 setosa
## 3      -999          3.2          1.3          0.2 setosa
## 4       4.6          3.1          1.5          0.2 setosa
## 5       5           3.6          1.4          0.2 setosa
## 6       5.4          3.9          1.7          0.4 setosa
## 7       4.6          3.4          1.4          0.3 setosa
## 8       5           3.4          1.5          0.2 setosa
## 9       4.4          2.9          1.4          0.2 setosa
## 10      4.9          3.1          1.5          0.1 setosa
## # i 140 more rows
```

Reading in multiple files

The `bind_rows()` function can be great for combining data.

recall that `modify` keeps the same data type (here, a list). We want a data frame instead.

See <https://www.opencasestudies.org/ocs-bp-vaping-case-study> for more information!

```
all_files_data <-  
  multiframe_data %>%  
  map_df(bind_rows, .id = "experiment")  
  
glimpse(all_files_data)
```

```
## Rows: 450  
## Columns: 6  
## $ experiment    <chr> "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1", "1...  
## $ Sepal.Length  <dbl> 5.1, 4.9, 4.7, 4.6, 5.0, 5.4, 4.6, 5.0, 4.4, 4.9, 5.4, 4...  
## $ Sepal.Width   <dbl> 3.5, 3.0, 3.2, 3.1, 3.6, 3.9, 3.4, 3.4, 2.9, 3.1, 3.7, 3...  
## $ Petal.Length  <dbl> 1.4, 1.4, 1.3, 1.5, 1.4, 1.7, 1.4, 1.5, 1.4, 1.5, 1.5, 1...  
## $ Petal.Width   <dbl> 0.2, 0.2, 0.2, 0.2, 0.2, 0.4, 0.3, 0.2, 0.2, 0.1, 0.2, 0...  
## $ Species       <chr> "setosa", "setosa", "setosa", "setosa", "setosa", "setosa...
```

Summary

- `function(x){ }` or `\(x){ }` denotes a function. You also commonly see `~.x` inside `across`
- `map_df` and `modify` **apply functions** to each element of an object. `map` returns a list, `modify` returns the same object type.
- The `purrr` package has other useful functional programming features
- lists can be great for storing iterative work.

<https://sisbid.github.io/Data-Wrangling/labs/functional-program-lab.Rmd>

Extra Slides

Using across with helpers to apply function to multiple columns

?tidyr_tidy_select

```
mtcars %>%  
  mutate(across(.cols = disp:wt, round)) %>%  
  head(2)
```

```
##  
##      mpg  cyl  disp  hp  drat  wt   qsec  vs  am  gear  carb  
## Mazda RX4      21   6  160 110    4   3  16.46  0   1    4    4  
## Mazda RX4 Wag  21   6  160 110    4   3  17.02  0   1    4    4
```

```
mtcars %>%  
  mutate(across(.cols = everything(), round))%>%  
  head(2)
```

```
##  
##      mpg  cyl  disp  hp  drat  wt   qsec  vs  am  gear  carb  
## Mazda RX4      21   6  160 110    4   3    16  0   1    4    4  
## Mazda RX4 Wag  21   6  160 110    4   3    17  0   1    4    4
```

Speed test! modify_if vs across

```
system.time(iris %>%  
  modify_if(is.factor, as.character))
```

```
##      user  system elapsed  
##    0.001    0.000    0.000
```

```
system.time(iris %>%  
  mutate(across(.cols = where(is.factor), as.character)))
```

```
##      user  system elapsed  
##    0.001    0.000    0.001
```

Alternatively we could have fixed the second file

Looks like it had a different delimiter First, separating by the `:`.

```
multifile_data[[2]] <-  
  separate(  
    multifile_data[[2]],  
    col = 1,  
    into = colnames(multifile_data[[1]]),  
    sep = ":"  
  )
```

```
## Warning: Expected 5 pieces. Missing pieces filled with `NA` in 150 rows [1, 2,  
## 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, ...].
```

```
head(multifile_data[[2]], 3)
```

```
## # A tibble: 3 × 5  
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species  
##   <chr>        <chr>        <chr>        <chr>        <chr>  
## 1 5.1         <NA>         <NA>         <NA>         <NA>  
## 2 4.9         <NA>         <NA>         <NA>         <NA>  
## 3 4.7         <NA>         <NA>         <NA>         <NA>
```

Fixing the second file

Second, making sure values are numeric.

```
multifile_data[[2]] <-  
  multifile_data[[2]] %>%  
  mutate(across(!Species, as.numeric))  
  
head(multifile_data[[2]], 3)
```

```
## # A tibble: 3 × 5  
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species  
##           <dbl>         <dbl>         <dbl>         <dbl> <chr>  
## 1           5.1           NA           NA           NA <NA>  
## 2           4.9           NA           NA           NA <NA>  
## 3           4.7           NA           NA           NA <NA>  
##  
##  
##
```